

Quantum Measurement, Minkowski Microstates and Matter Channels

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June 16, 2026

Abstract

Thermodynamic approaches to gravity, from Jacobson to Verlinde, leave unspecified the microscopic carriers of individual entropy increments. We show that the process of quantum measurement can naturally result in a gravitational microstate by developing an information-theoretic language of *matter-memory*, *handoff events*, and *spatial bits*, in which matter and horizon are complementary timelike and spacelike channels. Eliminating temperature between Landauer’s principle and the Unruh effect yields an acceleration-dependent bound on the energy associated with information. A single construction, a helicity-anticorrelated Bell state in flat Minkowski space, makes contact with Jacobson’s thermodynamic derivation of the Einstein equations, Van Raamsdonk’s entanglement-generated geometry, the BMS supertranslation structure at horizons.

1 Introduction

Jacobson derives the Einstein equations from entropy and temperature on local Rindler horizons [1]; Verlinde derives gravitational forces from entropy gradients on holographic screens [2]. Both frameworks operate at the level of bulk thermodynamics and leave open a prior question: *what physical process underlies each individual entropy increment?*

We will show how the answer can be quantum measurement by developing an information-theoretic language that connects “thrust” microstates, that can drive the Unruh effect [20], to the general quantum gravity program. We do this philosophically and with an explicit microstate construction which we then connect to current literature.

Before we dig in though, we first briefly outline our high level philosophical argument about how quantum measurement naturally connects to gravity. The general setup is pictured in the commutative¹ diagram in Figure 1, which we will walk through from the top right clockwise around the bottom to the top left. Quantum state reduction always happens due to an observation. Any observation is mediated by a physical interaction: information transfer necessarily involves an exchange of energy and momentum between the observed system and the detector, a discrete impulse. And then the equivalence principle implies that this impulse is locally indistinguishable from the effect of a gravitational field.

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¹This refactors the Diósi–Penrose postulate.

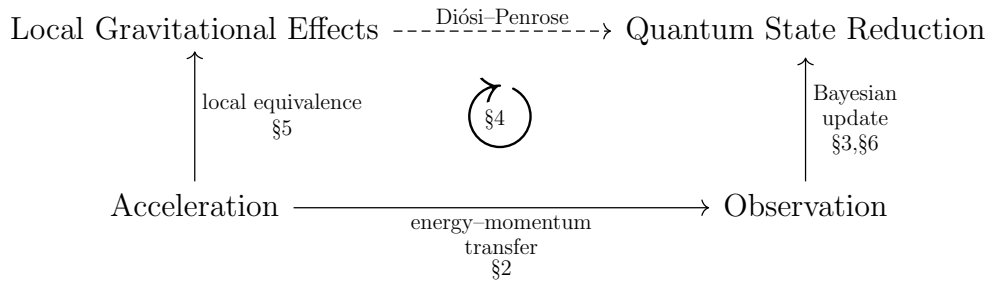


Figure 1: Solid arrows denote subjective grounding relations for a given observer: $A \rightarrow B$ means every instance of B is physically realized by an instance of A . This paper explores the solid arrows. The dashed arrow is the Diósi–Penrose postulate.

1.1 Roadmap

The paper develops each arrow of Figure 1 in turn, with Section 4 providing the information-theoretic framework that unifies them into a single physical event, one handoff between matter channels mediated by a massless quantum.

The **energy–momentum transfer** arrow is developed in Section 2: substituting the Unruh temperature into Landauer’s principle eliminates temperature entirely, yielding a purely kinematic bound on the energy cost of information acquisition under acceleration. The thrust quantum, the photon carrying the impulse, is the physical carrier of this bound.

The **Bayesian update** arrow is developed in Sections 3 and 6: the POVM-to-PVM transition selects a single non-thermal microstate from the Minkowski vacuum, the helicity outcome is recorded as a spatial bit in matter-memory, and the anti-correlation between right and left Rindler wedges is enforced by their common causal ancestry in the past wedge.

The **local equivalence** arrow is developed in Section 5: the absorbed photon sources an AS pp-wave at the Rindler horizon, the Raychaudhuri equation gives a discrete area increment $\Delta A = 8\pi G\hbar\omega$, and the event-by-event equality $S_{GHY}|\mathcal{H} = S_{\text{matter}}$ closes the consistency loop between geometry and information.

Section 4 is the conceptual core. It explains why these three arrows are aspects of one physical event: the handoff event deposits one quantum of action into matter-memory, records one classical bit, and imprints one Planck-area patch onto the horizon, all simultaneously. The Planck area ℓ_P^2 is the conversion factor between the geometric and informational descriptions.

2 The Unruh–Landauer Bound and Motivating Heuristics

A structuring choice runs throughout the paper. We work in the non-thermal global Minkowski description rather than the thermal Rindler description: thermality is a derived consequence of restricting to one wedge [7, 8], not a primitive, and individual microstates are non-thermal.

This section has two distinct parts. The Unruh–Landauer relation of Section 2.1 is an exact result that the construction layer inherits directly. substituting the Unruh temperature into Landauer’s bound eliminates temperature entirely, leaving a purely kinematic inequality depending only on acceleration. The subsequent derivations are explicitly Verlinde-style heuristics [2]: the same actors appear, acceleration, the Unruh relation, Newton’s laws, but thermality is eliminated rather than embraced. Where Ver-

linde deploys the thermal Rindler frame as the mechanism connecting entropy to force, the Unruh–Landauer relation eliminates temperature entirely, and Newton’s laws then recover standard black hole thermodynamics directly from the resulting kinematic bound. Thermality is a consequence of restricted causal access to one Rindler wedge [7, 8], not a primitive.

2.1 The Unruh–Landauer Relation

Landauer’s principle [11] establishes $k_B T$ as the natural energy scale of one distinguishable degree of freedom in a thermal environment at temperature T : any physical carrier representing ΔH nats must have at least this energy to be distinguishable from thermal noise,

$$\Delta E \geq k_B T \Delta H. \quad (1)$$

For an observer undergoing proper acceleration a , the Unruh effect associates a temperature

$$T = \frac{\hbar a}{2\pi k_B c} \quad (2)$$

with the local vacuum. Substituting into Landauer’s bound eliminates $k_B T$, yielding

$$\boxed{\Delta E \geq \frac{\hbar a}{2\pi c} \Delta H.} \quad (3)$$

Temperature has dropped out entirely: equation (3) is a temperature-independent lower bound on the energy of the physical carrier required to register one nat of information under proper acceleration a ; acceleration sets the minimum energy scale of the carrier. The full thermodynamic accounting, where the Landauer erasure cost appears, is deferred to Section ??.

The connection between Landauer’s principle and black hole thermodynamics has been noted previously. Cort  s and Liddle [12] showed that Hawking evaporation saturates the Landauer bound at the Hawking temperature, identifying the two results as expressions of the same process at the level of bulk evaporation. Related bounds on the energy cost of information acquisition in curved spacetime have been derived using the second law of thermodynamics applied to relativistic communication channels [13]. The present bound (3) differs in two respects: temperature is eliminated entirely by substituting the Unruh relation, yielding a purely kinematic bound depending only on acceleration, and the bound is applied to individual carrier quanta rather than to bulk evaporation.

This Unruh–Landauer bound is saturated for a photon at one bit when the wavelength equals the horizon distance up to a factor of $4\pi^2/\ln 2$:

$$\lambda_{\text{sat}} = \frac{4\pi^2}{\ln 2} \cdot \frac{c^2}{a} \sim d_{\mathcal{H}}, \quad (4)$$

where $d_{\mathcal{H}} = c^2/a$ is the proper distance to the Rindler horizon. The saturating photon is the fundamental mode of the causal diamond defined by the observer and the horizon, with wavelength commensurate with the horizon distance $d_{\mathcal{H}}$, too short to be super-horizon and too long to deposit more than one bit. This identifies the horizon distance as the natural length scale of a single-bit measurement event, and the Rindler horizon as the surface at which the Landauer bound is geometrically realized.

For realistic accelerations the saturation wavelength is enormous, at Earth surface gravity $a = 9.8 \text{ m s}^{-2}$, $\lambda_{\text{sat}} \approx 8$ light years, so the single-bit bound is practically always satisfied for terrestrial observers, and is only approached near astrophysical or Planck-scale accelerations.

2.2 Momentum Transfer and Acceleration

Note on method. Sections 2.2–2.5 are Verlinde-style dimensional analysis. They use Newtonian gravity and recover correct results at the Schwarzschild scale by the same coincidence exploited by Verlinde [2] and Bekenstein [14]: the Newtonian inverse-square law with a point mass reproduces the Schwarzschild geometry at the horizon scale, even though it formally breaks down there. This coincidence is specific to the spherically symmetric case; other matter distributions, including the bilocal source of Section 3, or the full Einstein equations, will in general give different results. The construction layer of Sections 3–5 makes no use of these heuristics. We make no attempt to disguise any of this.

In what follows, standard results are *reproduced as heuristic consequences* of the Unruh–Landauer relation and Newton’s laws. The energy transfer ΔE implied by the relation is carried by a physical quantum, matter or radiation, mediating the measurement interaction as a *thrust quantum*. For a relativistic carrier, the associated momentum transfer is

$$\Delta p = \frac{\Delta E}{c}. \quad (5)$$

A measurement event localized to a spatial scale r is bounded by a causal diamond with temporal extent $\Delta t = 2r/c$. This identification is the heuristic input from which the subsequent results follow, then

$$a = \frac{\Delta p}{m_d \Delta t} = \frac{\Delta E}{2m_d r}. \quad (6)$$

This is closely related to Verlinde’s entropic gravity [2]: Verlinde identifies holographic thrust on information screens with the origin of inertia itself, and Newton’s laws are a common ingredient in both constructions. The present approach differs in grounding each entropy increment in a discrete measurement event, the measurement impulse playing the role of Verlinde’s holographic thrust, rather than in a coarse-grained thermodynamic gradient.

2.3 The Schwarzschild Radius

We match the kinematic acceleration of the detector to the gravitational field of the thrust quantum using Newton’s laws. By $E = mc^2$, the energy quantum ΔE gravitates with effective mass $\Delta E/c^2$, producing a Newtonian gravitational acceleration

$$a = \frac{G\Delta E}{c^2 r^2} \quad (7)$$

at distance r from the detector. Equating this with the measurement-induced acceleration from equation (6),

$$\frac{\Delta E}{2m_d r} = \frac{G\Delta E}{c^2 r^2}. \quad (8)$$

The energy ΔE of the thrust quantum cancels entirely: the Schwarzschild radius is independent of the energy of the carrier. Solving for r :

$$r = \frac{2Gm_d}{c^2} \equiv r_s, \quad (9)$$

which is precisely the Schwarzschild radius of the detector mass m_d . The cancellation of ΔE is the physical content of the result: whatever energy the thrust quantum carries, the self-consistent horizon scale is set by the detector alone.

2.4 Surface Gravity

Evaluating the acceleration at $r = r_s$ gives the surface gravity of the detector:

$$\kappa = \frac{Gm_d}{r_s^2} = \frac{c^4}{4Gm_d}, \quad (10)$$

in agreement with the standard Schwarzschild result, as expected from the Verlinde coincidence. This reproduces the acceleration appearing in the Unruh temperature $T = \hbar\kappa/2\pi k_B c$, providing an internal consistency check on the heuristics.

2.5 Bekenstein–Hawking Entropy

Returning to the Unruh–Landauer relation (3) and substituting the surface gravity κ for a , justified by the equivalence principle: the local quantum field theory at the black hole horizon is the same as in the Rindler frame [15, 16], the entropy increment associated with an energy increment $\Delta E = c^2 \Delta m_d$ is

$$\Delta H \leq \frac{2\pi c}{\hbar\kappa} \Delta E = \frac{8\pi Gm_d}{\hbar c^3} \cdot \Delta m_d c^2. \quad (11)$$

Expressing this in terms of the change in Schwarzschild area $\Delta A = 32\pi G^2 m_d \Delta m_d / c^4$:

$$\Delta H \leq \frac{\Delta A}{4\ell_P^2}, \quad \ell_P^2 = \frac{\hbar G}{c^3}, \quad (12)$$

which is the Bekenstein–Hawking area law. Each measurement event contributes a discrete entropy increment proportional to the area imprinted on the horizon by the associated energy-momentum transfer.

This derivation parallels Jacob Bekenstein’s original argument [14], in which the entropy increase of a black hole is inferred from the minimal energy required to localize and drop a system across the horizon; here, the same structure emerges from the minimal energy cost of registering information at the horizon scale.

The two bounds are saturated simultaneously when $a = c^2/R$ — which is exactly the condition for being *at a horizon*. The horizon is therefore not just where thermodynamics happens to work out, it is the unique locus where the minimum cost of information acquisition equals the maximum information content of the region.

Substituting the horizon condition $a = c^2/R$ directly into the Unruh–Landauer bound (3) yields

$$\Delta H \leq \frac{2\pi R \Delta E}{\hbar c}, \quad (13)$$

which is precisely the Bekenstein universal entropy bound [17] on the entropy-to-energy ratio for a system of size R . The Unruh–Landauer relation and the Bekenstein bound are therefore the same inequality: the former is its acceleration-frame expression, the latter its position-space expression, and the Rindler horizon is the surface at which the two descriptions coincide. The covariant generalisation of the Bekenstein bound to arbitrary null hypersurfaces is the Bousso bound [18], which bounds entropy on any non-expanding null congruence by the area of its generating surface. The Raychaudhuri derivation of Section 5 operates precisely on such a congruence; the present construction can be read as a microstate-level realisation of the Bousso bound, with each handoff event contributing one Planck-area patch to the light-sheet.

Pesci [19] derived the covariant entropy bound from the Unruh temperature by applying the second law of thermodynamics across local Rindler horizons, the closest precursor to the present argument. The present bound equation (3) differs in two respects: temperature is eliminated entirely by substituting the Unruh relation into Landauer’s principle rather than retained as an input to the second law, yielding a purely kinematic bound depending only on acceleration; and the bound is applied to individual carrier quanta

rather than to a bulk thermodynamic flux. The identification of equation (3) with the Bekenstein bound [17] at the horizon condition $a = c^2/R$, and the reading of both as the same inequality in different frames, appears to be new in this form.

3 Bilocal Microstate Construction

From now on we use natural units $\hbar = c = 1$ unless otherwise specified; factors of $\hbar$ and c are restored where they aid physical interpretation.

The arguments of Section 2 motivate a heuristic relationship between information acquisition, acceleration, and gravitational effects, but leave open the question of what a single microstate looks like concretely. In standard treatments, the Unruh effect is derived by coupling a detector to a pre-existing acceleration and reading off a thermal response from the vacuum. In [20] this logic was inverted for a scalar field: a source placed in the past injects non-thermal entangled excitations into the field, and the observer’s accelerated response was shown to be driveable by these source-induced excitations. We extend that construction here to the spin-1 electromagnetic field, upgrading from scalar to helicity-carrying Bell states and making the single-bit structure of each measurement event explicit.

A *Minkowski microstate* is, in this paper, a single explicit quantum event in flat Minkowski space, a photon pair emitted by a physical source, detected by two accelerated observers, that realises one discrete entropy increment of the kind that Jacobson and Verlinde require at the thermodynamic level. We make no claim that this is the unique such construction; other configurations, with back-reaction, asymmetric detector placement, or different source geometries, would also qualify. The construction below is simply the smallest one we could exhibit that is clean enough to track consistently through the geometric, thermodynamic, information-theoretic, and effective field theory frameworks simultaneously.

3.1 A Source and Two Observers

Three massive bodies are placed at spacetime positions $(t, z) = (0, 1)$, $(0, -1)$, and $(-1, 0)$, named $\mathcal{R}$ – *Right*, $\mathcal{L}$ – *Left*, and $\mathcal{P}$ – *Past* respectively. $\mathcal{P}$ serves as the common causal ancestor of the accelerations of $\mathcal{R}$ and $\mathcal{L}$ (see Figure 2). $\mathcal{P}$ emits an entangled pair of spin-1 massless photons with momenta $\pm k$ and Rindler frequency $\omega = \omega_k$. The momenta are perfectly balanced by construction: the two photons carry spatial momenta $+k\hat{z}$ and $-k\hat{z}$ respectively, so the total linear momentum vanishes, $k + (-k) = 0$. The angular momentum vanishes term by term: in each helicity assignment $(\lambda_R, \lambda_L) = (+1, -1)$ and $(-1, +1)$, the total helicity is $\lambda_R + \lambda_L = (+1) + (-1) = 0$ and $(-1) + (+1) = 0$ respectively, and the symmetric Bell state superposition (26) preserves this since both terms contribute zero independently. The source therefore exhibits no net back-reaction in either linear or angular momentum. The two photons propagate into the left and right Rindler wedges, where $\mathcal{L}$ and $\mathcal{R}$ detect them respectively via absorption, the momentum impulse of which accelerates each detector into its respective Rindler wedge; each observer thereby registers a helicity eigenvalue and acquires one classical bit of information. The emission event at $\mathcal{P}$ naturally defines null coordinates $u = t - z + 1$ and $v = t + z + 1$, centered so that $\mathcal{P}$ lies at $u = v = 0$, the right-moving photon propagates along $u = 0$, and the left-moving photon propagates along $v = 0$.

The three-party structure $\mathcal{P}$ – $\mathcal{R}$ – $\mathcal{L}$ realises a form of tripartite entanglement with a natural interior reconstruction interpretation. Neither $\mathcal{R}$ nor $\mathcal{L}$ alone can distinguish a correlated Bell-state emission at $\mathcal{P}$ from an uncorrelated thermal source: each observer’s reduced density matrix is thermal, exactly as in the standard Unruh effect. Together, their two anticorrelated bits reveal the correlation structure of the emission event, identifying it as a specific Bell state at a specific frequency, and providing a microstate-level analogue

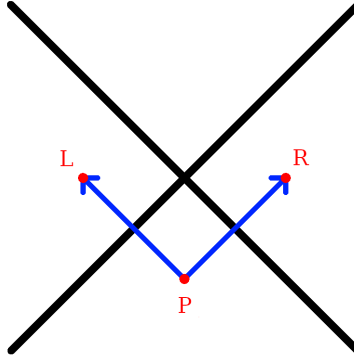


Figure 2: A source in the past, $\mathcal{P}$, emits an entangled photon pair that accelerates $\mathcal{R}$ into the right Rindler wedge and $\mathcal{L}$ into the left Rindler wedge. The two null directions of propagation define the null coordinates $u = t - z + 1$ and $v = t + z + 1$, with the right-moving photon propagating along $u = 0$ and the left-moving photon along $v = 0$; these are the z^- and z^+ axes of the double-null coordinate system of Section 3.4.

of black hole interior reconstruction [21, 22], where the interior is recovered as classical correlations in the joint record of complementary exterior observers.

3.2 PVM Localization and the Elimination of Coarse-Graining

The Unruh mode plays a double role in this construction. As a QFT object it is a coefficient in a mode expansion, a solution to the wave equation analytically continued from the Rindler wedge into all of Minkowski space. But once $\mathcal{P}$ is specified as a real physical source via the squeezing interaction (20), the mode function acquires a second interpretation: it is a probability amplitude in the Born rule sense, encoding where $\mathcal{R}$ is likely to detect the emitted quantum. This is the inversion of the standard Unruh–DeWitt logic. Rather than coupling a detector to a pre-existing acceleration and reading off a thermal distribution by averaging over all Rindler frequencies, we front-load the logic at $\mathcal{P}$: the source fixes the frequency ω_k and the mode function then specifies the probability amplitude for detection.

Thermality, in this picture, is not a primitive property of the vacuum seen by an accelerating observer but a consequence of residual ignorance, about the precise location of $\mathcal{R}$ within the wedge, and about the spatial origin of the detected excitation, that the present construction resolves by specifying the common causal ancestry of $\mathcal{P}$, $\mathcal{R}$, and $\mathcal{L}$ explicitly. The ignorance is epistemic, not ontic: the global Minkowski state remains pure throughout, and thermality is what a single-wedge observer sees when causal access to $\mathcal{P}$ ’s emission event is unavailable.

This inversion has a geometric consequence that runs through the entire paper: it is precisely what allows the construction to live in flat Minkowski space rather than requiring an AdS/CFT dual. In the standard holographic picture, geometry is primary and entanglement is read off it, Van Raamsdonk’s argument [3] proceeds from the entanglement entropy of a boundary state to the geometry of the bulk ER bridge.

Here the direction of explanation is reversed: the emission event at $\mathcal{P}$ is primary, and the geometry, the AS pp-wave, the horizon area increment, the BMS supertranslation, is the *ontic* output of the subsequent measurement events at $\mathcal{R}$ and $\mathcal{L}$. The geometry is a real physical change; what is epistemic is the assignment of that change to a particular observer’s microstate record. These are distinct, and keeping them distinct is what allows the construction to live in flat Minkowski space without a holographic dual.

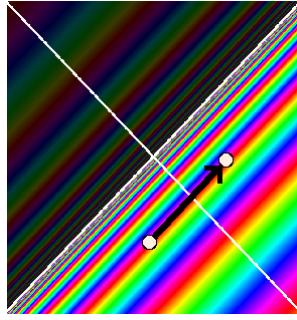


Figure 3: The points $\mathcal{P}$ and $\mathcal{R}$ are marked with white dots, connected by the null geodesic (black arrow). The rainbow coloring shows the Unruh mode phase structure across Minkowski space. A PVM projection onto this null line selects a single frequency from the thermal superposition, replacing the full delocalized mode with the localized non-thermal microstate of equation (26).

Longitudinal localization. The primary localization is in the longitudinal (t, z) plane, which is the setting of Figure 3 and of the parabolic cylinder function analysis of [20]. In that work, a Minkowski mode φ_q^* multiplied pointwise by a Gaussian shaping function $N_{\mu, \sigma}$ of width σ in (t, z) , then paired via the Klein–Gordon inner product with a Rindler mode r_k , yields an overlap expressible in terms of parabolic cylinder functions $D_\nu(-\zeta)$. The Gaussian could equivalently have been applied to the Rindler side; what matters is that σ controls the degree of longitudinal localization of the source-detector pair. The Stokes phenomenon in the asymptotics of $D_\nu(-\zeta)$ as σ varies drives a transition between two physically distinct regimes: $\sigma \rightarrow \infty$ gives the thermal regime where the $e^{-\frac{1}{4}\zeta^2}$ branch dominates, and $\sigma \rightarrow 0$ gives the localized non-thermal regime where the $e^{+\frac{1}{4}\zeta^2}$ branch dominates, with the Stokes line at $\arg \zeta = \pi/4$ marking the boundary between them.

In the present paper the same Gaussian shaping function is reread as a Kraus operator acting on the quantum state.

$$K_\sigma = \int e^{-u^2/2\sigma^2} |u\rangle\langle u| du, \quad (14)$$

where $u = t - z + 1$ is the null coordinate² defined in Section 5. The operator K_σ together with its complement

$$K_\sigma^\perp = \int \sqrt{1 - e^{-u^2/\sigma^2}} |u\rangle\langle u| du, \quad (15)$$

satisfying $K_\sigma^\dagger K_\sigma + (K_\sigma^\perp)^\dagger K_\sigma^\perp = \mathbf{1}$ by construction, defines a one-parameter POVM family; the Stokes transition between its two asymptotic regimes is illustrated in Figure 4. Specifying $\mathcal{R} = (0, 1)$ and $\mathcal{P} = (-1, 0)$ is precisely the $\sigma \rightarrow 0$ limit: the Kraus operator sharpens to a PVM projector onto the unique null geodesic connecting them, selecting a single definite frequency ω_k from what would otherwise be a thermal superposition over all Rindler frequencies. This is visible directly in Figure 3: the rainbow phase structure of the Unruh mode concentrates onto the null line between $\mathcal{P}$ and $\mathcal{R}$, and the PVM projection selects that line. The Stokes transition of [20] is therefore not merely a field-theoretic regularization result but the analytic signature of the POVM-to-PVM transition: the same mathematical object is being read in two complementary languages, field theory and measurement theory, with the Stokes line marking the boundary between them. The helicity label λ enters the longitudinal mode function as a spectator index and does not affect the u -dependence of the Kraus operator (14); the Stokes transition analysis of [20] therefore applies independently to each helicity sector without modification.

²We use ζ for the Stokes variable of [20] to avoid confusion with the longitudinal spatial coordinate z .

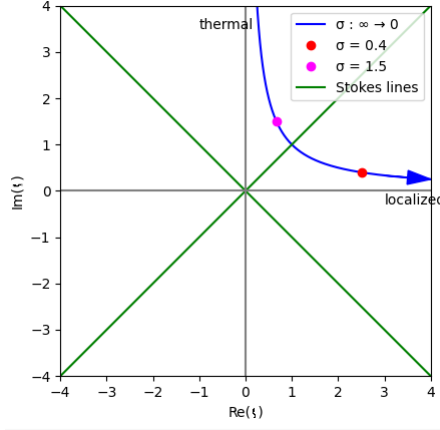


Figure 4: Trajectory of $\zeta = iq\sigma + \mu/\sigma$ in the complex plane as σ interpolates between the two limits of the Kraus family (14). The thermal limit $\sigma \rightarrow \infty$ corresponds to ζ on the positive imaginary axis; the localized PVM limit $\sigma \rightarrow 0$ corresponds to ζ on the positive real axis. The green Stokes lines at $\arg \zeta = \pi/4$ mark the boundary between the two asymptotic regimes of the parabolic cylinder function $D_\nu(-\zeta)$, and therefore the boundary between the POVM and PVM descriptions of the longitudinal localization. Figure from [20].

Transverse localization. The transverse (x, y) directions require a separate and independent localization. A Rindler plane wave at frequency ω is a coherent superposition over all possible source-detector pairs $(\mathcal{P}', \mathcal{R}')$ connected by null geodesics at that frequency, uniform in the transverse plane. The longitudinal and transverse localizations decouple because the wave equation in double-null coordinates separates: the u -dependence and the (x, y) -dependence of the mode function enter independently, so the two Kraus projections can be applied in either order without interference. Specifying $\mathcal{P} = (-1, 0)$ and $\mathcal{R} = (0, 1)$ performs a second PVM projection, now in the transverse directions, selecting the unique null geodesic and resolving the remaining coarse-graining over transverse position. The transverse shaping is controlled by an independent width $\sigma_\perp$, which we treat as a free parameter in Section 3.7 and whose physical consequences are developed in Section 5.

The two widths σ and $\sigma_\perp$ control physically distinct aspects of the construction and need not be sent to zero together. The longitudinal σ governs the POVM-to-PVM transition and the thermal-to-non-thermal transition analyzed via the Stokes phenomenon; Section 3 works entirely in the $\sigma \rightarrow 0$ PVM limit. The transverse $\sigma_\perp$ controls the spatial profile of the horizon perturbation and the BMS mode content: $\sigma_\perp \rightarrow \infty$ produces a planar impulsive wave exciting a single BMS supertranslation mode, while $\sigma_\perp \rightarrow 0$ recovers the Aichelburg–Sexl pp-wave with logarithmic shift $\Delta v \propto -\ln r^2$ exciting the full BMS spectrum. The area increment $\Delta A = 8\pi G\hbar\omega$ and the event-by-event equality $S_{GHY}|\mathcal{H} = S_{\text{matter}}$, which serve as the geometric consistency checks of the framework, are independent of $\sigma_\perp$, fixed by energy conservation alone.

The microstate. The result of both projections is an individually non-thermal microstate carrying a definite Rindler frequency ω and a definite longitudinal position on the null geodesic from $\mathcal{P}$ to $\mathcal{R}$. Thermality is not a property of any single microstate but of the ensemble; it emerges when one models the detector response within a wedge as a sum over all Rindler modes, as in the standard Unruh–DeWitt treatment [15, 23], and is a consequence of that mode-averaging rather than a premise of the construction.

The individual non-thermal microstate is what the PVM projection onto the $\mathcal{P}$ – $\mathcal{R}$ null geodesic selects; the thermal ensemble is what remains when neither coarse-graining is resolved and $\mathcal{L}$ is traced out.

A note on the $\sigma \rightarrow 0$ idealization: throughout this paper the PVM limit is understood as the idealization of a physical construction that lives at small but finite σ . At exactly $\sigma = 0$ the Kraus operator (14) becomes a projector onto a set of measure zero in the continuum $|u\rangle$ basis and is no longer normalizable in the strict sense; similarly, the AS longitudinal shift $\Delta v \propto -\ln r^2$ diverges on axis at $r = 0$ in the strict localized limit. Both divergences are regularized by keeping $\sigma > 0$, as carried out explicitly for the geometric side in Appendix ???. The parameter σ therefore plays a unified role: it is simultaneously the width of the Kraus operator in the longitudinal direction, the transverse profile width of the photon mode, and the regularization scale of the AS singularity, all controlled by the same physical quantity, the photon wavelength $\sigma \sim c/\omega$. All sharp statements about individual microstates should be understood as holding in the limit $\sigma \rightarrow 0^+$ rather than at exactly $\sigma = 0$.

3.3 Helicity Structure and PCT Symmetry

We now upgrade this construction from a spin-0 scalar field in [20] to the spin-1 electromagnetic field with a superposition of helicity pairs in a Bell state, introducing one unknown binary degree of freedom, the helicity outcome, undetermined until measured, which is the single bit responsible for the remaining thermality in the ensemble. The photon carries one quantum of action $\hbar$ and energy $E = \hbar\omega$, identifying the thrust quantum of Section 2 with a single photon whose polarization outcome, upon detection, is the classical bit acquired by each observer. The Bell state is the natural completion of the PCT symmetry between wedges, which by the Bisognano–Wichmann theorem [8] is implemented by the modular conjugation operator J mapping the right wedge algebra to the left wedge algebra, and as we discuss in Section 5, the photons imprint a definite geometric signature onto the horizon.

The anticorrelated helicity assignment traces to the PCT symmetry implemented by the Bisognano–Wichmann modular conjugation J , which maps the right-wedge algebra to the left-wedge algebra and simultaneously flips $\lambda \rightarrow -\lambda$; the Bell state (26) is the natural state in this momentum sector invariant under that operation. The common algebraic origin of this anticorrelation in the Lorentz algebra decomposition $\mathfrak{so}(3, 1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$, developed in Section 4.5.

3.4 Double-Null Gauge

The emission event naturally defines two null directions: the right-moving photon has wave-vector $k_R^\mu = (\omega, 0, 0, k)$ and the left-moving photon has wave-vector $k_L^\mu = (\omega, 0, 0, -k)$, with equal and opposite spatial momenta. These two null directions are precisely the z^+ and z^- axes of double-null coordinates

$$z^\pm = \frac{1}{\sqrt{2}}(t \pm z) = \frac{1}{\sqrt{2}}(u - 1 \pm (v - 1)), \quad (16)$$

with transverse coordinates $\mathbf{z}_\perp = (x, y)$. Note that $z^\pm$ are centered on the Minkowski origin $(t, z) = (0, 0)$ rather than on $\mathcal{P}$; the coordinates u, v defined in Section 3 are the $\mathcal{P}$ -centered versions and are used throughout the remainder of the paper. Working in this frame we impose the symmetric double-null gauge condition

$$A_+(z) = 0, \quad A_-(z) = 0, \quad (17)$$

so that the gauge field is supported entirely on the transverse components $\mathbf{A}_\perp = (A^x, A^y)$. This gauge respects the PCT symmetry that swaps the two wedges ($z^+ \leftrightarrow z^-$) simultaneously with $(R \leftrightarrow L)$, and therefore treats both observers symmetrically by construction.

The residual gauge freedom is fixed by requiring $\partial_i A^i = 0$ in the transverse plane, leaving precisely the two physical helicity degrees of freedom $\epsilon_{(\lambda)}^i$ with $\lambda = \pm 1$; consistency with Maxwell's equations and the absence of over-determination are verified in Appendix ??.

The lack of manifest four-dimensional covariance in equation (17) is a consequence of aligning the gauge with the pair of physical null directions selected by the source, rather than an arbitrary choice. The full Lorentz group is broken to the little group preserving the pair $\{k_R^\mu, k_L^\mu\}$, which is the two-dimensional Euclidean group $ISO(2)$ acting on the transverse plane. This is precisely the little group of massless particles, and the two helicity eigenstates $\lambda = \pm 1$ are its irreducible representations. All physical results are independent of the gauge choice: the helicity eigenvalues $\lambda = \pm 1$ are representations of the massless little group $ISO(2)$ and are therefore Poincaré-covariant objects independent of how the gauge is fixed, and the consistency of the double-null gauge with Maxwell's equations and the correct degree-of-freedom count is verified explicitly in Appendix ??.

3.5 The Bilocal Source

In double-null gauge the electromagnetic field is characterised by its transverse components alone. The bilocal driving source is symmetrized over both helicity assignments,

$$J^{ij}(x, y) = g \left(\epsilon_{(-1)}^{i*} \epsilon_{(+1)}^j + \epsilon_{(+1)}^{i*} \epsilon_{(-1)}^j \right) f_L(x) f_R(y), \quad (18)$$

where f_L and f_R are normalised mode profiles supported on the left and right wedges respectively, and $\epsilon_{(\lambda)}^i$ are the circular polarization vectors

$$\epsilon_{(+1)}^i = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad \epsilon_{(-1)}^i = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \quad (19)$$

The two terms in (18) are related by the PCT transformation in the sense of Bisognano–Wichmann [8], which maps the right wedge algebra to the left wedge algebra and simultaneously flips $\lambda \rightarrow -\lambda$.

More precisely, the modular conjugation J of Bisognano–Wichmann maps a right-wedge creation operator $a_{k,(\lambda)}^{R\dagger}$ to a left-wedge creation operator $a_{-k,(-\lambda)}^{L\dagger}$, implementing PCT between wedges. The Bell state (26) is a natural two-photon state at fixed frequency ω and fixed momentum pair $(k, -k)$, invariant under J : within this momentum sector, the anticorrelated helicity assignment and the symmetric superposition are not choices but consequences of requiring J -invariance, since J flips both helicity and wedge simultaneously, leaving no room for a relative phase or helicity-correlated alternative. The bilocal source (18) is therefore a natural source, consistent with the algebraic symmetry of the Rindler vacuum, and the simplest one we could construct with the required properties.

The symmetrization is consistent with PCT symmetry and angular momentum conservation at the emission vertex. Other source geometries are possible; this one is chosen for its simplicity and the transparency of its connection to the algebraic structure of the Rindler vacuum. The argument has three steps. First, a scalar source carries zero angular momentum, so the emitted pair must satisfy $\lambda_R + \lambda_L = 0$, which forces the two helicities to be anticorrelated: the only possibilities are $(\lambda_R, \lambda_L) = (+1, -1)$ or $(-1, +1)$. Second, the Bisognano–Wichmann modular conjugation J maps the first assignment to the second and vice versa, so the unique J -invariant state in this sector is their symmetric superposition with relative phase fixed by J -invariance. Third, any other relative phase or asymmetric combination would break the PCT symmetry between wedges, which is a symmetry of the Minkowski vacuum and must therefore be preserved by the source. The total angular momentum vanishes in both terms independently.

The bilocal source selects a specific microstate from the Minkowski vacuum by a PVM projection onto the $n = 1$ sector at fixed ω and fixed helicity pair $(\lambda, -\lambda)$, in exact analogy with the scalar construction of [20]. No perturbative expansion in a coupling

constant is required: the helicity-anticorrelated photon pair is already present in the vacuum decomposition (24), and the bilocal source selects it via the PVM projection of Section 3.2. The operator content of this selection is

$$a_{-k,(-1)}^{L\dagger} a_{k,(+1)}^{R\dagger} + a_{-k,(+1)}^{L\dagger} a_{k,(-1)}^{R\dagger} + \text{h.c.}, \quad (20)$$

where the two creation operator pairs correspond to the two PCT-related helicity assignments, each carrying total angular momentum zero, consistent with emission from a scalar source at $\mathcal{P}$. This is the vector-field analogue of the scalar squeezing interaction [24], with the PCT symmetry between wedges made explicit; the connection to the quantum optics squeezing literature is structural rather than perturbative.

Each photon carries energy $E = \hbar\omega$ and one quantum of action $\hbar$, identifying the thrust quantum of Section 2 with a concrete physical object: the minimum actionable event is precisely a single photon. Its polarization, undetermined until measured, is the quantum bit whose outcome constitutes the classical record acquired by each observer upon detection.

The Unruh–Landauer bound is saturated not at the level of an individual photon but at the level of the source rate. The acceleration a does not fix the frequency ω of each selected quantum; rather, it determines the rate $\dot{N}$ at which the bilocal source selects entangled pairs in order to implement that acceleration, with the total power $\dot{N}\hbar\omega$ matching the Unruh thermal flux. Each individual photon carries exactly one quantum of action $\hbar$ and one binary helicity outcome, one bit in the sense of one binary alternative carrying $\ln 2$ nats, and these are paired by the construction independently of a . The Landauer bound then operates as a consistency condition on the ensemble: the minimum energy cost per bit acquired, summed over the rate of acquisition, equals the power required to sustain the acceleration.

3.6 Polarization Bogoliubov Transformation

We now show that the polarization labels survive the Rindler–Unruh mode decomposition, so that the thermal ensemble retains its helicity structure and the anticorrelation of the emitted pair descends unambiguously to the level of individual Rindler microstates.

In the transverse plane, the electromagnetic field in double-null gauge decomposes into two independent scalar-like sectors labelled by helicity $\lambda = \pm 1$. For each sector, the right-wedge field expands as

$$A_\lambda^i(z) = \int_0^\infty \frac{d\omega}{2\pi} \frac{1}{\sqrt{2\omega}} \epsilon_{(\lambda)}^i \left[b_{\omega,\lambda}^R u_\omega^R(z) + b_{\omega,\lambda}^{R\dagger} u_\omega^{R*}(z) \right], \quad (21)$$

where $u_\omega^R(z)$ are the right-wedge Rindler modes at Rindler frequency ω , and $b_{\omega,\lambda}^R, b_{\omega,\lambda}^{R\dagger}$ are the corresponding annihilation and creation operators satisfying

$$\left[b_{\omega,\lambda}^R, b_{\omega',\lambda'}^{R\dagger} \right] = \delta(\omega - \omega') \delta_{\lambda\lambda'}. \quad (22)$$

An identical expansion holds in the left wedge with $R \rightarrow L$ and $\lambda \rightarrow -\lambda$, reflecting the anticorrelated helicity assignment. Because each helicity sector is independently a scalar-like sector, the Bogoliubov transformation relating Minkowski and Rindler modes takes the same form in each sector. For helicity λ at Rindler frequency ω , the Unruh modes are

$$c_{\omega,\lambda} = \cosh \theta_\omega b_{\omega,\lambda}^R - \sinh \theta_\omega b_{\omega,-\lambda}^{L\dagger}, \quad (23)$$

where $\tanh \theta_\omega = e^{-\pi\omega c/a}$ encodes the Unruh distribution at acceleration a . The helicity flip $\lambda \rightarrow -\lambda$ on the left-wedge operator is not an independent assumption but a direct consequence of the Bisognano–Wichmann modular conjugation J , which acts as $J b_{\omega,\lambda}^{R\dagger} J^{-1} = b_{-\omega,-\lambda}^{L\dagger}$ and therefore requires the left-wedge partner of a right-wedge mode

of helicity λ to carry helicity $-\lambda$; the PCT symmetry between wedges is thereby reflected in the superposition over both helicity assignments in the Bell state. The transformation (23) is the standard Bogoliubov relation for a massless field [15, 25], now carrying a helicity label throughout. The Minkowski vacuum thermofield double state, in the Rindler basis reads, for each helicity sector,

$$|0\rangle_M^{(\lambda)} = \frac{1}{\cosh \theta_\omega} \sum_{n=0}^{\infty} \tanh^n \theta_\omega |n_{\omega,\lambda}\rangle_R |n_{\omega,-\lambda}\rangle_L, \quad (24)$$

and the full vacuum is the product over all ω and both helicities:

$$|0\rangle_M = \prod_{\omega,\lambda} |0\rangle_M^{(\lambda)}. \quad (25)$$

Equation (24) makes explicit that the Minkowski vacuum is a sum over entangled helicity-anticorrelated pairs at each Rindler frequency. The bilocal source specifies a physical emission event: exactly one pair, at fixed ω , with fixed helicity assignment. The PVM projection of Section 3.2 identifies this event with the $n = 1$ term in the vacuum decomposition (24) at that ω and helicity pair. The higher n terms are not suppressed by any weak-coupling assumption; they correspond to different physical events, two-pair emissions, three-pair emissions, and so on, that this source simply does not describe. The selection is prescriptive, not perturbative. The resulting single microstate is

$$|\Psi_\omega\rangle_M = \frac{1}{\sqrt{2}} (|1_{\omega,+1}\rangle_R |1_{\omega,-1}\rangle_L + |1_{\omega,-1}\rangle_R |1_{\omega,+1}\rangle_L). \quad (26)$$

$\mathcal{R}$'s and $\mathcal{L}$'s helicity outcomes are undetermined until measurement, at which point they are perfectly anticorrelated by the Bell state structure: each observer acquires one classical bit, and the two bits are complements of one another.

The key structural result is that the helicity label λ is conserved under the Bogoliubov transformation: it enters equation (23) as a spectator index and exits unchanged on the Unruh operators $c_{\omega,\lambda}$. Thermalality, when it arises from coarse-graining over the ensemble of microstates (26), is therefore helicity-resolved: each helicity sector thermalises independently, and the anticorrelation between wedges is preserved at every level of the construction.

3.7 Two Geometric Regimes

The microstate (26) admits two complementary geometric descriptions depending on the transverse localisation of the photon mode, corresponding to the two limits of the transverse Kraus family

$$K_{\sigma_\perp} = \int e^{-r^2/2\sigma_\perp^2} |\mathbf{x}_\perp\rangle \langle \mathbf{x}_\perp| d^2 x_\perp, \quad (27)$$

parametrized by $\sigma_\perp$, the direct transverse analogue of the longitudinal Kraus family (14) with the null coordinate u replaced by the transverse radial coordinate $r = |\mathbf{x}_\perp|$. The bilocal source fixes a definite Rindler frequency ω but imposes no constraint on transverse momentum. A mode with zero transverse momentum is uniform across the horizon cross-section; a wave packet of characteristic width $\sigma \sim c/\omega$ is localised within one wavelength of the photon axis. These two limits correspond to geometrically distinct pp-waves at the Rindler horizon, both analysed in Section 5:

- **Delocalized limit** ($\sigma_\perp \rightarrow \infty$, POVM $\rightarrow \mathbf{1}$): the stress tensor is uniform across the horizon, producing a *planar impulsive wave* with longitudinal shift $\Delta v \propto -r^2$, which is a supertranslation in the BMS sense.

- **Localized limit** ($\sigma_\perp \rightarrow 0$, POVM $\rightarrow$ PVM): the stress tensor concentrates near the beam axis, recovering the Aichelburg–Sextl pp-wave with logarithmic shift $\Delta v \propto -\ln r^2$. All BMS supertranslation modes are excited.

Both limits produce the same total horizon area increment $\Delta A = 8\pi G\hbar\omega$, as shown in Section 5 and Appendix ???. The physical reason is that the Raychaudhuri equation integrates the null energy T_{uu} over the entire transverse horizon cross-section: regardless of how that energy is distributed transversely, the integral $\int T_{uu} du d^2x_\perp = \hbar\omega$ is fixed by energy conservation alone, so the total area increment is insensitive to the transverse profile. The difference between the two limits lies entirely in the spatial distribution of that increment across the horizon and the corresponding BMS mode content excited by each profile.

4 Matter as a Substrate

We propose that matter is the fundamental substrate of spacetime information processing by reinterpreting quantum mechanics in an information-theoretic way: proper time emerges from discrete action quanta accumulated along worldlines, spatial geometry emerges from handoff events at horizons, and the Planck area ℓ_P^2 is the conversion factor between the two. This gives a Lorentz-invariant alternative to approaches that discretize geometry directly, discreteness lives in the matter sector via $\hbar$, and geometry inherits it through the covariant Raychaudhuri equation. The following subsections develop this in stages:

1. **Matter as a timelike channel** (Section 4.1): a massive particle carries phase at the Compton frequency, setting its clock rate and maximum information throughput.
2. **The horizon as a spacelike channel** (Section 4.2): the horizon is a separating surface through which two timelike matter channels communicate with ℓ_P^2 as the conversion factor.
3. **The handoff event** (Section 4.3): The communication, an elementary massless carrier, transfers action and information between matter-memories.

We then discuss particle types and types of interactions (Section 4.4), and finish up with a way to understand Dirac’s electron as a Compton clock (Section 4.5) connecting our framework to the Standard Model via Penrose’s zig-zag.

Our approach sidesteps a prior question about quantum gravity: *how can one quantize gravity without directly quantizing spacetime or its curvature degrees of freedom?* Common approaches posit Planck-scale cells [26] or discrete spacetime events [27], but Lorentz invariance is recovered only statistically in such constructions rather than at the level of individual configurations [28]. We instead recover discreteness from matter: space and time are properties of matter itself, defined only up to the resolution set by the action $S/\hbar$, which is Lorentz invariant by construction.

4.1 Matter as a Timelike Channel

A massive particle carries phase along its worldline at a rate set by its rest energy. The Compton frequency $f = mc^2/\hbar$ is the natural clock rate of a mass m : it is the rate at which the phase $e^{iS/\hbar}$ completes one cycle. This is the content of Bremermann’s computational limit [29] which describes the maximum clock rate of any physical system with bounded energy. We associate a discrete event with one zero crossing of the phase $e^{iS/\hbar}$ in the path integral and we use the information-theoretic term *sample* throughout. The sampling resolution is limited by the Heisenberg uncertainty $\Delta E \Delta t \sim \hbar$, which plays the role of the Nyquist limit for the matter channel [30]: It would require additional energy $\Delta E > mc^2$ to probe the sub-Compton phase.

The matter substrate gives rise to a channel capacity in the Shannon sense: for a system of mass m with internal Hilbert space dimension d , the maximum rate of information flow is

$$C = \underbrace{\frac{mc^2}{\hbar}}_{\text{bandwidth}} \times \underbrace{\log_2(d)}_{\text{dynamic range}} \quad (\text{bits per second}), \quad (28)$$

where the bandwidth is the Bremermann sampling rate and the dynamic range is the information content per sample. Both quantities are extensive under composition:

$$\frac{mc^2}{\hbar} = \frac{(m_1 + m_2)c^2}{\hbar} = \frac{m_1 c^2}{\hbar} + \frac{m_2 c^2}{\hbar}, \quad (29)$$

$$\log_2(d) = \log_2(d_1 d_2) = \log_2(d_1) + \log_2(d_2), \quad (30)$$

consistent with their interpretation as properties of matter that compose naturally under union.

The signal being sampled at the Bremermann rate is the internal quantum state³ of the particle, its spin, helicity, and other quantum numbers, and the dynamic range is $\log_2$ of the dimension of the internal Hilbert space. The channel capacity (28) sets the rate at which matter can accumulate and process information, under the assumption that each Bremermann sample accesses the full d -dimensional internal Hilbert space.

4.2 The Horizon as a Spacelike Channel

The three fundamental constants c , $\hbar$, and G play precise roles in the channel picture. The speed of light c sets the causal propagation speed of signals. The reduced Planck constant $\hbar$ sets the resolution of the timelike channel via the Compton frequency $mc^2/\hbar$. The gravitational constant G , through the Schwarzschild radius $r_s = 2Gm/c^2$, sets the maximum energy storable in a spatial region without gravitational collapse. Their combination $\ell_P^2 = \hbar G/c^3$ is the Planck area, the minimal spatial region per independent quantum of action.

The horizon enters the channel picture in a double role. As a *channel*, it is the surface through which causal connection is established: the photon crosses it and that crossing is the handoff event, converting quantum entanglement between the wedges into a classical record in matter-memory. As a *barrier*, it is the surface at which this conversion makes the opposite wedge causally irrelevant to the receiving observer's future evolution. The disconnection between wedges is therefore not a pre-existing feature of the geometry that the photon happens to cross; it is a consequence of the handoff itself. Before the crossing $\mathcal{R}$ and $\mathcal{L}$ share a global pure state; after it, $\mathcal{L}$ is inaccessible to $\mathcal{R}$ precisely because the entanglement has been consumed in producing the classical record.

4.3 The Handoff Event

The photon crossing the horizon is a handoff between the timelike matter channels of $\mathcal{R}$ and $\mathcal{L}$, mediated by the spacelike horizon channel. To that end we define:

Definition (Handoff Event). A *handoff event* is a horizon-crossing interaction that transfers action and information between matter-memories.

In the explicit bilocal construction of Section 3, this takes the following specific form:

³For composite systems the internal Hilbert space includes additional degrees of freedom beyond spin and helicity: relative positions and momenta of constituents, vibrational modes, and entanglement between subsystems. How these contribute to the accessible dimension per Bremermann sample, and whether they generate additional spatial structure analogous to the spatial bit of Section 6, are questions the present framework motivates but does not address.

- (i) a massless quantum of Rindler frequency ω and helicity $\lambda = \pm 1$, selected by the PVM projection of Section 3.2, transferring one quantum of action $\hbar$ across $\mathcal{H}$;
- (ii) the receiving timelike channel incrementing its Bremermann sampling rate by $\Delta f = \omega/2\pi$.
- (iii) the helicity outcome λ recorded as one classical bit in the receiving channel's matter-memory; and
- (iv) the horizon area incrementing by $\Delta A = 8\pi G\hbar\omega$.

Property (iv) follows from the Raychaudhuri equation applied to the null congruence at $\mathcal{H}$ and is derived in Section 5; it is listed here to make the geometric content of the handoff explicit. Whether properties (i)–(iii) generalize beyond the photon case, to massive carriers, higher-spin fields, or POVM rather than PVM projections, is a question the construction motivates but does not answer.

Each handoff event has exactly two irreducible constituents: one quantum of action $\hbar$ written into matter-memory, incrementing the Bremermann sampling rate, and one bit of classical information, the helicity outcome. These are not independent: they are the energy $\hbar\omega$ and angular momentum $\pm\hbar$ of a single photon, two properties of the same physical object. With this definition in hand, the equality $S_{GHY}|\mathcal{H} = S_{\text{matter}}$ of Section 5 becomes a counting statement: the GHY boundary term sums $\hbar\omega$ over handoff events because that is what the term geometrically records, one contribution per event.

4.4 Types of Particles and Interactions

The distinction between massless and massive degrees of freedom is operational in the channel picture: massless quanta transmit information between matter substrates as hand-off carriers, while massive systems retain it as matter-memory. A massless particle has no rest frame, no proper time, and carries a single quantum of action between emission and absorption; it does not sample and therefore carries no persistent memory. Its role is precisely that of a pure information carrier effecting a handoff between the timelike channel of one massive body and that of another.

From a given observer's perspective, interactions fall into three classes. **Absorption** produces a persistent classical record: a quantum of action is deposited and information is written. **Emission** is the reverse: stored information is released and the record is lost. **Reflection** is absorption followed by emission; whether it constitutes an information-bearing interaction depends on whether a persistent record survives. The ideal mirror of Section ?? is precisely the limiting case of reflection with no persistent record: it absorbs the photon, temporarily stores the helicity outcome, and re-emits without retaining a copy, making the interaction thermodynamically reversible at the cost of paying the Landauer erasure price at re-emission.

These categories are observer-relative in the same sense as quantum state reduction: just as state reduction is a Bayesian update for a given observer grounded in their causal access to the measurement outcome, the classification of an interaction as absorption, emission, or reflection depends on whether the observer in question retains a persistent record of the outcome. The same physical process may be information-bearing for one observer and purely external for another, depending on whether a record is written into their matter-memory.

4.5 The Tick-Tock Mechanism

The handoff event of Section 4.3 has a natural analogue along a timelike worldline. Penrose's zig-zag construction [31] represents a massive fermion as two massless Weyl spinors of opposite helicity, coupled at each vertex by the Higgs field and alternating in a zig-zag pattern; we term this the tick-tock mechanism to emphasise the alternating rhythm

of handoff events rather than the geometry of the path. The parallel with the bilocal construction is structural: the Higgs field plays the role of $\mathcal{P}$, the two Weyl spinors play the role of the two photons propagating to $\mathcal{R}$ and $\mathcal{L}$, and each vertex is a handoff event depositing one quantum of action and producing one sample of proper time rather than one increment of horizon area. We present this as a suggestive analogy rather than a derived result.

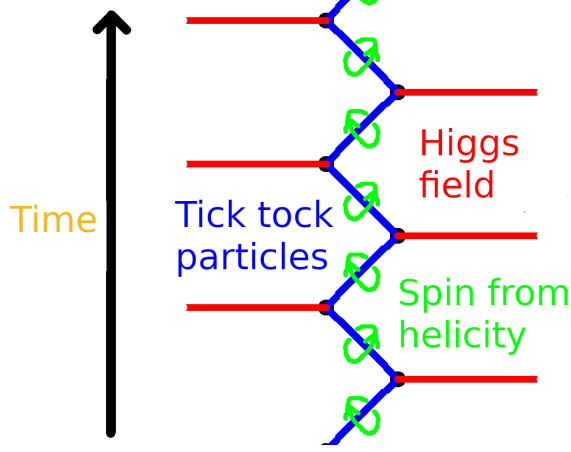


Figure 5: The tick-tock mechanism. A massive particle (blue) alternates between right- and left-helicity massless segments, coupled at each vertex by the Higgs field (red). Spin emerges from the alternating helicity flip at each vertex (green loops), visible as the two $\mathfrak{su}(2)$ factors of $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$ cycling between handoff events. Each vertex is a handoff event: one quantum of action deposited, one helicity outcome recorded, one sample of proper time produced.

The anticorrelated helicity assignment in both constructions has a common algebraic origin. The Bisognano–Wichmann modular conjugation J acts on right-wedge photon creation operators as

$$J a_{k,\lambda}^{R\dagger} J^{-1} = a_{-k,-\lambda}^{L\dagger}, \quad (31)$$

flipping helicity and reversing the spacelike separation direction $R \leftrightarrow L$. The PCT operator Θ acts on a right-handed Weyl spinor as

$$\Theta \psi_R(t, \mathbf{x}) \Theta^{-1} \propto \psi_L(-t, -\mathbf{x}), \quad (32)$$

flipping helicity and reversing the timelike separation direction $t \rightarrow -t$. In both cases the operation is the same at the representation-theoretic level: exchange of the two $\mathfrak{su}(2)$ factors of $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$. The operators J and Θ act on different Hilbert spaces; what they share is that both select anticorrelated helicities as the unique PCT-invariant configuration. between

4.6 Relation to Existing Programs

The matter-substrate framework of this section makes contact with two independent programs in quantum foundations that approach emergent time from complementary directions.

The Page–Wootters mechanism [32, 33] derives time evolution relationally: a global static state $|\Psi\rangle \in \mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{sys}}$ satisfying a Wheeler–DeWitt-like constraint yields emergent dynamics when conditioned on clock eigenstates $|t\rangle$. The mechanism is exact but

leaves the physical identity of the clock system unspecified. The present framework supplies that identity: the matter substrate is the clock, with phase $e^{iS/\hbar}$ advancing at the Bremermann rate $mc^2/\hbar$. The standard mechanism assumes a continuous clock time t ; the present construction replaces this with a discrete sequence of handoff events, each acting as a PVM conditioning step that advances the Bremermann count by one quantum of action. Proper time is relational in exactly the Page–Wootters sense, but the clock states are physical rather than abstract.

The modular flow perspective of Connes and Rovelli [34] approaches the same problem algebraically. In a generally covariant quantum theory the Hamiltonian vanishes on physical states and no preferred time exists; Connes and Rovelli show that a state ω on a von Neumann algebra $\mathcal{M}$ nonetheless generates a canonical one-parameter automorphism group, the modular flow σ_t^ω , via Tomita–Takesaki theory. This flow satisfies the KMS condition and is thermodynamically indistinguishable from thermal time evolution. The Bisognano–Wichmann theorem [8], already central to the PCT structure of Section 3, identifies this modular flow concretely with Rindler time evolution for the vacuum restricted to one wedge. The connection to the present framework is therefore already implicit in the construction: the horizon $\mathcal{H}$ is precisely the algebraic cutoff that Connes and Rovelli require, restricting the observable algebra to the right wedge and generating the modular flow as a consequence. What the present framework adds is a discrete substrate beneath that flow: the continuous modular parameter is the thermodynamic limit of discrete Bremermann samples.

The two programs are thus not independent supports for emergent time but complementary descriptions of the same structure at different levels: Page–Wootters gives the kinematic conditioning on individual clock states, Connes–Rovelli gives the thermodynamic flow that emerges when those states are coarse-grained across a horizon. The matter substrate and the handoff event are the microscopic objects that both programs require but neither specifies.

Verlinde’s entropic gravity [2] operates at the thermodynamic level of the same horizon structure; the present framework supplies the microstate layer beneath it, with each handoff event realising one entropy increment of the kind Verlinde’s screens accumulate in bulk. The Wigner’s friend problem [35] and the Frauchiger–Renner theorem [36] identify the multi-observer problem as formally sharp; the vocabulary introduced here, spatial bits, PVM projections, handoff events, matter-memory, is the natural language in which that problem must be posed when the observers are causally separated by a horizon, a point developed in Section 6.

5 pp-Wave Geometry and Action

We introduce coordinates for the photon traveling from $\mathcal{P}$ $(-1, 0)$ to $\mathcal{R}$ $(0, 1)$ (there is a similar story for the left wedge). We work in the null coordinates u, v defined in Section 3, where the photon propagates along the null plane $u = 0$, z is the longitudinal direction, and (x, y) are transverse coordinates. Then $(u, v) = (0, 0)$ is $\mathcal{P}$ ’s location, $(u, v) = (0, 2)$ is $\mathcal{R}$ ’s location, and $(u, v) = (0, 1)$ is the intersection of the photon with the past right Rindler horizon.

5.1 The AS pp-Wave and Its Effects

Test particles at fixed transverse positions (x, y) experience two effects as the shock front crosses $u = 0$:

1. **Longitudinal shift.** The v coordinate is displaced by

$$\Delta v \propto -\ln r^2, \quad r^2 = x^2 + y^2. \quad (33)$$

This shift is lightlike: no proper time elapses for a particle crossing the shock, but it is displaced along the null direction. Near the axis ($r \rightarrow 0$) the shift diverges, encoding the concentration of energy along the photon's trajectory.

2. **Transverse focusing.** The gradient of the longitudinal shift produces a transverse velocity

$$\Delta \mathbf{v}_\perp \propto \nabla \ln r \propto \frac{1}{r}, \quad (34)$$

directed toward the axis. Test particles are pulled toward the centre of the wave, with the kick strength growing as $1/r$ toward the axis.

5.2 The Longitudinal Shift Makes Space

The longitudinal shift deserves closer attention, because it carries a meaning beyond a mere coordinate displacement. The photon emitted by $\mathcal{P}$ propagates along the null direction; as it crosses the Rindler horizon, the AS shift displaces the future light cone of every test particle it passes. For the photon traveling into the right wedge, the shift pushes the right wedge's geometry in the $+v$ direction; for the entangled partner traveling into the left wedge, the shift pushes the left wedge in the $+u$ direction. The two wedges are displaced *away from each other* along the longitudinal null direction. See Figure 6.

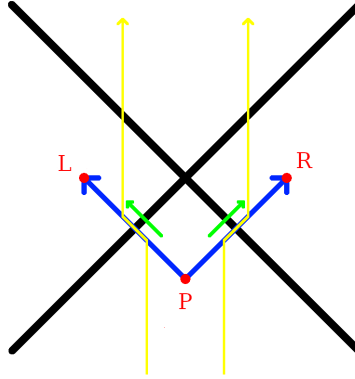


Figure 6: The green arrows show the longitudinal shift that occurs where the photons cross the horizon. Test particles, shown in yellow, are pushed outwards away from $z = 0$ due to the creation of space along the null lines originating from $\mathcal{P}$.

At the intersection of the shock front $u = 0$ with the past right Rindler horizon at $(u, v) = (0, 1)$, the transverse (x, y) plane is identified with the horizon area elements $dA = dx dy$. The longitudinal shift $\Delta v \propto -\ln r^2$ displaces the horizon outward in the v direction, pushing the right wedge geometry away from the horizon. The entangled partner photon produces a symmetric shift in the u direction at the left Rindler horizon, pushing the left wedge away from its horizon. The two wedges are therefore displaced away from their respective horizons in opposite null directions.

5.3 Transverse Focusing Increments Horizon Area

The area increment Δ is independent of the transverse localization of the photon. In the delocalized limit the stress tensor is uniform across the horizon and produces a planar shift $\Delta v \propto -r^2$; in the localized limit it concentrates near the beam axis and produces the AS logarithmic shift $\Delta v \propto -\ln r^2$. Both are supertranslations in the BMS sense and both give the same area increment, as the Raychaudhuri calculation below shows. The one-parameter family of Gaussian transverse profiles interpolating between these two limits, parametrized by width σ , is the geometric counterpart of the POVM-to-PVM transition

of Section 3.2: the spatial distribution of the area increment sharpens from uniform to concentrated as $\sigma \rightarrow 0$, while the total ΔA remains fixed by energy conservation throughout.

We derive the area increment using the Raychaudhuri equation on the null congruence generating the Rindler horizon, following Jacobson [1] and consistent with the generalized second law for Rindler horizons proved by Wall [37], but applied here to a single microstate rather than a thermodynamic or ensemble average.

For a null congruence with generators k^μ and expansion θ , the Raychaudhuri equation gives

$$\frac{d\theta}{du} = -\frac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - 8\pi G T_{\mu\nu}k^\mu k^\nu. \quad (35)$$

Integrating over the horizon patch in the linearised regime and using $k^\mu = \delta_u^\mu$, the area change is

$$\Delta A = 8\pi G \int T_{uu} du d^2x_\perp. \quad (36)$$

The present construction provides a natural regularization of the strict AS point-source limit: the photon is a normalizable mode of the electromagnetic field in double-null gauge, with a smooth transverse profile set by the wavelength $\sim c/\omega$. The electromagnetic stress-energy T_{uu} is well-defined throughout, integrates to $\hbar\omega$ over the transverse plane by energy conservation, and recovers the strict AS limit as $\omega \rightarrow \infty$ with the logarithmic divergence regulated at the scale c/ω .

Substituting into equation (36) gives

$$\boxed{\Delta A = 8\pi G \hbar\omega = 8\pi G \Delta E.} \quad (37)$$

The horizon area grows by one discrete, calculable increment per photon absorbed, microstate by microstate, without reference to bulk thermodynamic averages. This is property (iv) of the handoff definition of Section 4.3 verified geometrically: the area increment $\Delta A = 8\pi G \hbar\omega$ is the geometric expression of one quantum of action $\hbar$ deposited into matter-memory by one measurement event, with the Planck area ℓ_P^2 as the conversion factor between the two descriptions.

Here the null generators of the horizon are taken to be affinely parametrised as $k^\mu = (\partial/\partial u)^\mu$ in double-null coordinates, so that $\int T_{uu} du d^2x_\perp = \hbar\omega$ is the total null energy of the photon. This differs from the Killing-normalised convention by a factor of the surface gravity $\kappa = a$; the two are related by $(\partial/\partial u)^\mu = a(\partial/\partial \eta)^\mu$ at the horizon, consistent with the standard result $\Delta A = 8\pi G E/\kappa$ when $E = \hbar\omega$ is the Killing energy.

The longitudinal shift $\Delta v \propto -\ln r^2$ produced by each AS pp-wave at the Rindler horizon is a supertranslation in the sense of Bondi–Metzner–Sachs (BMS) symmetry. Hawking, Perry, and Strominger [5] showed explicitly that a black hole is supertranslated by the passage of an asymmetric shockwave, and the AS longitudinal shift is precisely the linearised supertranslation field on the horizon. Carlip [4] demonstrated that the near-horizon diffeomorphisms of a generic black hole are enhanced to a BMS_3 algebra whose central charge determines the Bekenstein–Hawking entropy, but his program identifies the symmetry without specifying the physical process that excites individual modes. The bilocal construction supplies that process: each measurement event produces one AS pp-wave and deposits one supertranslation at the Rindler horizon. The shockwave S-matrix program of 't Hooft [38] established that the longitudinal AS shift is the fundamental dynamical ingredient for horizon information exchange at the level of the full S-matrix; the present construction realises the same shift at the level of a single microstate.

5.4 Action and the Gravitational Boundary

The equality derived below is an internal consistency check on the microstate identification, not a novel result claimed in isolation. The thermodynamic interpretation of

the GHY boundary term on null surfaces is established by Chakraborty and Padmanabhan [40]; the broader program of Padmanabhan [41] identifies gravitational boundary terms as the fundamental degrees of freedom at the ensemble level. What the present construction contributes is not the identity $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ but its realisation microstate by microstate: if individual quantum measurement events genuinely are the microstates underlying each area increment, the geometric count and the informational count must agree event by event. The calculation below verifies that they do, closing the consistency loop.

Each absorption event transfers null energy $\hbar\omega_i$ across the horizon. The matter action at the horizon is defined as the null-surface integral of the stress-energy, evaluated over the horizon generators:

$$S_{\text{matter}} = \sum_i \int T_{uu}^{(i)} du d^2x_{\perp} = \sum_i \hbar\omega_i, \quad (38)$$

where the second equality follows from energy conservation applied to each photon. This is the null-surface analog of the worldline action $\int E d\tau$: on a null surface the affine parameter u plays the role that proper time plays on a timelike worldline, and the integral $\int T_{uu} du$ plays the role of $\int E d\tau$.

The two are not the same integral. After the absorption event, the detector's worldline action continues to accumulate as $\hbar\omega_i \Delta\tau$ for all subsequent proper time $\Delta\tau$, encoding the Bremermann memory of the event. The equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ holds for the null-surface version of the matter action, evaluated at the moment of horizon crossing; it is a statement about the handoff event itself, not about the subsequent worldline evolution. The Rindler proper time η and the affine null parameter u are related by $u \propto e^{-a\eta}$, and it is this exponential relation, not a simple proportionality, that is responsible for the Unruh temperature and that distinguishes the null-surface action from the worldline action. This is the matter action counted sample by sample: one quantum of action per event, one event per emitted pair.

The same events have a geometric description. By eq. (37), each absorption sample contributes a horizon area increment $\Delta A_i = 8\pi G \hbar\omega_i$, so the total area increment summed over all events is

$$\sum_i \Delta A_i = 8\pi G \sum_i \hbar\omega_i = 8\pi G S_{\text{matter}}. \quad (39)$$

This geometric sum has a precise identification in the gravitational action. On a stationary Rindler horizon the unperturbed expansion of the null generators vanishes, $\theta = 0$, so the Gibbons–Hawking–York boundary term [42] is sourced entirely by the AS pp-wave perturbations:

$$S_{GHY}|_{\mathcal{H}} = \frac{1}{8\pi G} \int_{\mathcal{H}} \theta d^2x_{\perp} du = \frac{1}{8\pi G} \sum_i \Delta A_i = \frac{1}{8\pi G} \cdot 8\pi G \sum_i \hbar\omega_i = \sum_i \hbar\omega_i = S_{\text{matter}}. \quad (40)$$

Chakraborty and Padmanabhan [40] showed that the gravitational boundary term on null surfaces has a thermodynamic interpretation as a heat density Ts of the null surface, establishing $S_{GHY}|_{\mathcal{H}} = \text{heat content}$ at the level of thermodynamic averages. The present result realises the same identity at the level of individual microstates, with each contribution to S_{GHY} sourced by a specific quantum measurement event rather than a bulk thermal average.

The matter action and the gravitational boundary action are therefore equal, event by event, on the stationary Rindler horizon. This is not a bulk result: the bulk Einstein–Hilbert action vanishes on-shell, and it is the boundary term that carries the physical content. The microstate construction identifies what that boundary content is: each contribution to S_{GHY} corresponds to one quantum of action deposited into matter-memory by one measurement event, precisely the handoff event of Section 4.3. The event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ is therefore the geometric consistency check on the

information-theoretic framework of Section 4: the Raychaudhuri equation and the hand-off definition count the same events in two languages, and they agree.

The equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ suggests a stronger reading, which we flag as speculative but worth stating. The bulk Einstein–Hilbert action vanishes on-shell, so it is the boundary term alone that carries the gravitational degrees of freedom. If the boundary term counts matter action quanta event by event, as equation (40) shows, then the geometry at the horizon is not an independent quantity that happens to equal the matter action: it *is* the matter action, expressed in geometric language. The Planck area $\ell_P^2 = \hbar G/c^3$ is the conversion factor between the two descriptions, relating one quantum of action $\hbar$ to one Planck-area patch on the horizon. In this reading the metric near the horizon is not a smooth field that matter happens to perturb; it is a coarse-grained count of the action quanta that have passed through the horizon surface, with the Bekenstein–Hawking area law as the statement that these two counts agree. We do not claim to establish this here; it is a direction the present construction motivates and that we leave for future work. Whether this identification extends from the boundary into the bulk, that is, whether the Ricci scalar R can be understood as an action-quanta density with ℓ_P^2 as the conversion factor, is the central open question.

This result is consistent with and extends programs of Jacobson [1] and Padmanabhan [41], which argue that the gravitational boundary term rather than the bulk Einstein–Hilbert action is the fundamental gravitational degree of freedom. Those programs establish this at the level of thermodynamic averages. The present construction realizes the same identification at the level of individual microstates, with each boundary contribution sourced by a specific quantum measurement event rather than a bulk average. The standard result emerges from summing over the ensemble; the microstate picture supplies what the thermodynamic programs leave open: the physical process underlying each boundary contribution.

The equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ is structurally related to the Ryu–Takayanagi formula [43], which identifies the entanglement entropy of a boundary region with the area of a minimal bulk surface in AdS: $S_{\text{ent}} = A/4G\hbar$. Three differences sharpen the comparison. First, the present result holds in flat Minkowski space on a Rindler horizon rather than on a minimal surface in AdS, and requires no holographic dual. Second, it is an equality of actions rather than of entropies, with the Planck area ℓ_P^2 as the conversion factor between the two sides rather than as a fixed coefficient. Third, it holds event by event rather than as a thermodynamic average over a thermal ensemble: the AS pp-wave sourced by a single photon produces a single calculable area increment, and summing over events recovers the thermodynamic result. Whether $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ is the flat-space Rindler limit of the Ryu–Takayanagi formula, or whether the two results are independent derivations of the same underlying holographic principle from orthogonal directions, is a natural question for future work.

The classical focusing theorem used above to derive $\Delta A = 8\pi G\hbar\omega$ has a quantum generalisation, the Quantum Null Energy Condition (QNEC) [44, 45], which replaces the null energy condition with a bound involving the second derivative of von Neumann entropy of the outgoing field state along the null generators. The QNEC is the natural framework for extending the present construction beyond pure microstates: the classical focusing theorem suffices here because the bilocal construction maintains a globally pure state throughout, with no need to trace out either wedge. At the boundary of EFT validity, where the single-wedge Rindler description requires tracing out $\mathcal{L}$ and the global pure state gives way to a mixed thermal state, the QNEC becomes the operative inequality; that it saturates precisely at $\omega = \omega_{\text{sat}}$ is consistent with the six equivalent conditions of Proposition ??, and we leave its precise role at that boundary for future work.

6 Entanglement as Spatial Information

The two geometric regimes of Section 3.7 carry complementary entanglement interpretations. In the delocalized limit, the planar impulsive wave uniformly displaces the entire horizon outward in the $+v$ direction (right wedge) and $-v$ direction (left wedge): the two wedges are pushed apart globally, and the geometric connection between them is generated by the entanglement of the full thermofield double state (24). This is the ensemble-level version of Van Raamsdonk’s argument [3]: the delocalized photon produces a global deformation of the geometry connecting the two wedges via a BMS supertranslation at the horizon connecting the two wedges. In the localized limit, the AS longitudinal shift is concentrated in a cylindrical neighbourhood of the photon trajectory: the two wedges are displaced apart locally, microstate by microstate, and the geometric connection is built up event by event from individual measurement interactions. The full Van Raamsdonk picture is the sum of these local events over the ensemble; the bilocal construction makes each contributing event explicit.

Van Raamsdonk [3] showed, in the AdS/CFT setting, that the entanglement entropy between two halves of a thermofield double state controls the geometry connecting them: reducing entanglement elongates the ER bridge until the geometry disconnects entirely. The vacuum decomposition (24) is precisely the thermofield double state [46], now in flat Minkowski space with helicity labels carried through. The AS longitudinal shifts of Section 5 are its event-by-event geometric output: each measurement event displaces the two wedges apart in opposite null directions, creating new coordinate space between them.

A potential sign puzzle deserves immediate attention. Van Raamsdonk’s argument implies that a maximally entangled state (such as the Bell state (26)) should produce maximal geometric *connection* between the two sides, whereas the AS shifts push the wedges *apart*. The resolution is that outward displacement in the present construction is not disconnection but creation of interior volume: the two wedges are not separating topologically but making room for the geometry between them, consistent with the positive longitudinal shift generating new Minkowski space rather than a change in causal structure. This is the event-by-event realisation of Van Raamsdonk’s entanglement-generated geometry. Three features go beyond his argument: the setting is flat space without an AdS/CFT dual; the result holds microstate by microstate rather than at the level of entanglement entropy; and the source $\mathcal{P}$ is explicit, providing the physical mechanism that generates the entanglement.

The bilocal source prepares an anticorrelated Bell state (26). We consider the simplest case: both $\mathcal{R}$ and $\mathcal{L}$ measure in the same basis and both obtain a definite outcome, which we label H_+ or H_- . In this case their outcomes are perfectly anticorrelated by the Bell state structure, independently of which basis is chosen. The question

“who observed H_+ ?”

then has a binary answer: $\mathcal{L}$ (left wedge) or $\mathcal{R}$ (right wedge). This is a *spatial bit*, a bit that encodes not a polarization state but a location in space. The spatial bit is where the observer-relativity of the construction becomes acute. From $\mathcal{R}$ ’s perspective the answer to “*who observed H_+ ?*” is a definite classical fact, recorded in matter-memory. From $\mathcal{P}$ ’s perspective before any measurement, the answer is indeterminate: the Bell state assigns no definite location to H_+ . From the full Minkowski perspective after both measurements, the two records are perfectly correlated but the question of *when* the spatial fact became definite has no observer-independent answer. A PVM by $\mathcal{R}$ on a Bell state does not reveal a pre-existing spatial fact; it *creates* the spatial bit as a classical record in $\mathcal{R}$ ’s matter-memory. This is the same PVM projection as Section 3.2, now read informationally rather than geometrically: the projection that selects the localized microstate from the thermofield double is the same event as the projection that creates the spatial bit in matter-memory.

Whether that creation is an objective event or a relational one is a question the present construction does not need to resolve in full generality. Within this construction, the spatial bit is relational but not arbitrary: its creation is grounded in the common causal ancestry of $\mathcal{P}$, which enforces the anticorrelation between $\mathcal{R}$'s and $\mathcal{L}$'s records as a consequence of shared causal history rather than an objective collapse.

Each measurement event has exactly two irreducible constituents: one quantum of action $\hbar$ written persistently into the detector's matter-memory as a series of samples, and one bit of polarization registered as a classical outcome. The entanglement between $\mathcal{R}$ and $\mathcal{L}$ is consumed in producing these two records, an increase in sampling rate and one bit at each wedge. What remains after measurement is not the entanglement itself but its classical residue: the perfect anticorrelation between the two bits, and the correlated pair of samples written into the matter on each side of the horizon. The spatial bit is precisely the polarization bit now read as a location rather than a spin state: the same bit that records *what* was observed also records *where*, because in this construction the two wedges carry opposite helicities by design. This double duty is forced: the anticorrelated helicity assignment of the bilocal source makes the two wedges distinguishable by construction, so the spatial bit is already latent in the source geometry before any measurement occurs.

The information is carried by the matter of the detectors, not by the horizon. As established in Section 4, the horizon is a boundary that makes the left/right distinction sharp; it is the matter on each side that accumulates the samples and holds the classical record. $\mathcal{R}$'s detector has absorbed one photon, increased its sampling rate, and registered polarization H_+/H_- . $\mathcal{L}$'s detector has absorbed one photon, increased its sampling rate, and registered polarization H_-/H_+ . The two records are perfectly complementary by the anticorrelated structure of the bilocal source, and their complementarity is encoded in matter and is reflected in geometry.

6.1 Emergent Space and the Thermofield Double

The Unruh vacuum is the thermofield double state of the left and right Rindler wedges [47]: equation (24) is precisely this state, with the helicity label λ carried through. The bilocal source selects one term from this sum at fixed ω and fixed helicity pair, yielding the single microstate (26). The correlated AS longitudinal shifts in opposite null directions are the geometric output of this microstate: a linearized perturbation of flat Minkowski space that displaces the two wedges away from their respective horizons, creating new coordinate space between them.

The connection between the two wedges requires no exotic geometry: $\mathcal{R}$ and $\mathcal{L}$ are correlated because they share a common causal past in $\mathcal{P}$, which is ordinary retarded causality. Non-traversability is equally automatic, no signal can pass directly from $\mathcal{L}$ to $\mathcal{R}$ because the causal structure of Minkowski space forbids it, matching the no-communication theorem on the EPR side without additional assumptions. The ER=EPR correspondence, in this flat-space setting, reduces to the statement that entangled systems with a common causal ancestor generate correlated geometric displacements, a consequence of energy-momentum conservation at $\mathcal{P}$ rather than a postulate about wormhole topology. The wormhole picture of ER=EPR may be what this flat-space causal structure looks like in AdS, when the source $\mathcal{P}$ is not explicitly available and the connection between wedges must be encoded in the geometry itself.

7 Conclusion

We have proposed quantum measurement events as the elementary units of spacetime information processing and shown this is consistent with geometric, thermodynamic, and effective field theory descriptions of the horizon. The central claim, that matter is the timelike substrate of emergent proper time, the horizon is the spacelike surface through

which handoff events generate emergent space, and the Planck area ℓ_P^2 is the conversion factor between the two descriptions, gives a specific, tangible, Lorentz-invariant alternative to approaches that quantize geometry directly. The event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ is the consistency check that validates this framework, not the result itself. The construction makes contact with three established programs.

Jacobson’s thermodynamic derivation of the Einstein equations [1] requires a physical process underlying each local entropy increment; the bilocal measurement event is that process, identified explicitly rather than postulated. Van Raamsdonk’s entanglement-generates-geometry argument [3], formulated at the level of reduced density matrices in AdS/CFT, is realised here in flat space microstate by microstate, with the AS longitudinal shift as the concrete geometric output of each entanglement event. The BMS soft-theorem-memory triangle [5, 6] identifies the symmetry structure at the horizon; the bilocal construction populates that structure with a concrete physical process: each measurement event produces an AS longitudinal shift that is a supertranslation in the BMS sense.

The geometric consistency check is the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$: the gravitational boundary action and the matter action are the same quantity counted in two languages, with the Planck area ℓ_P^2 as the conversion factor. This follows from a single temperature-eliminated Unruh–Landauer bound, from which standard black hole thermodynamics emerges via Newton’s laws, and is consistent across thermodynamic, geometric, information-theoretic, and effective field theory descriptions of the horizon. Matter and horizon are orthogonal information channels: matter the timelike substrate of emergent proper time, the horizon the spacelike surface through which PVM projections onto null geodesics generate emergent space.

A related but distinct novelty concerns the epistemic status of the horizon and its entropy. In the standard treatment, Unruh thermality is observer-dependent but objective within each observer’s description: the partial trace over the inaccessible wedge produces a genuinely mixed state, not a state of ignorance about an underlying pure microstate. The present construction inverts this. The global Minkowski state is pure throughout; the thermal state that $\mathcal{R}$ sees is a consequence of two coarse-grainings simultaneously, ignorance of $\mathcal{P}$ ’s emission event and ignorance of $\mathcal{R}$ ’s location within the wedge, both of which the PVM projection eliminates explicitly. Thermality is therefore epistemic in a precise sense: it is what remains when the source is unspecified, not an objective feature of the vacuum that geometry imposes on accelerating observers. The horizon plays a corresponding double role. As a causal surface it is ontic, signals cannot pass from $\mathcal{L}$ to $\mathcal{R}$. But as an epistemic boundary it is the surface at which $\mathcal{R}$ ’s single-wedge EFT description and the global Minkowski description become incompatible, the geometric expression of the point at which ignorance becomes irreducible for a single observer.

Horizon entropy, in this reading, is not a property of the geometry alone but a measure of the epistemic gap between these two levels of description, with the Planck area ℓ_P^2 as the conversion factor between the ontic geometric count and the epistemic informational count. The ontic causal surface and the epistemic entropy surface coincide at the horizon because causal inaccessibility and informational irreducibility are the same condition, approached from the spacetime and information-theoretic sides respectively.

This is what the inversion from $\mathcal{R}$ ’s passive thermal reading to $\mathcal{P}$ ’s active emission event achieves, and it is precisely what allows the construction to live in flat Minkowski space: no holographic dual is needed because the geometry is the output of measurement events, not their prerequisite.

The observer-relativity of the microstate construction is a feature rather than a liability. The Bekenstein–Hawking area and horizon entropy are objective, observer-independent quantities; the microstates identified here, PVM projections by a specific Rindler observer $\mathcal{R}$, recording a spatial bit in a specific matter-memory, are observer-relative. These are consistent because observer-relative microstate counts are grounded in the common

causal ancestry of $\mathcal{P}$: the anticorrelation between $\mathcal{R}$'s and $\mathcal{L}$'s records is enforced by shared causal history, and the observer-independent area law is the statement that these observer-relative counts are mutually consistent. The remaining open question is not how to resolve the observer-relativity but how to extend this causal grounding to a complete multi-observer formalism, a question the information-theoretic vocabulary introduced here, spatial bits, PVM projections, handoff events, matter-memory, is equipped to pose precisely.

The Frauchiger–Renner theorem [36] shows this is not merely a philosophical gap but a formal one, with measurable consequences; in the gravitational context, with causally separated observers and non-commuting measurement algebras across a horizon, the problem is sharper still. The information-theoretic vocabulary introduced here, spatial bits, PVM projections, handoff events, matter-memory, and the Planck area as conversion factor, is what is needed to state the problem precisely, even if it does not resolve it.

The construction takes no stand on the ontology of the wavefunction. The results, the area increment $\Delta A = 8\pi G\hbar\omega$, the predicted statistics, and the Landauer cost of the erasure step, are the same whether the PVM projection is read as a Bayesian update, a relational fact, or a branching event. The interpretational commitment lies entirely in what one takes the classical record in matter-memory to mean. The relation to the Diósi–Penrose objective collapse program [9, 10] is precise on this point: that program postulates observer-independent gravitational collapse of the wavefunction, whereas the present construction treats state reduction as a Bayesian update for a given observer, with the dashed arrow of Figure 1 the only relation we do not adopt.

Whether the event-by-event equality $S_{GHY}|\mathcal{H} = S_{\text{matter}}$ extends from the boundary into the bulk, whether the helicity structure of the emergent space constitutes a topological invariant, whether the framework admits falsifiable consequences distinguishing it from standard thermodynamic treatments, and whether the flat-space construction presented here is the Rindler limit of a more general statement about arbitrary horizons, remain open questions for future work.

Acknowledgements

Portions of this work were developed in dialogue with Claude (Anthropic). The author takes full responsibility for all claims.

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